

In [10]: `!pip install numpy`

Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: numpy in c:\programdata\anaconda3\lib\site-packages (2.3.5)

Create a NumPy array with values [10, 20, 30, 40]

In [57]: `import numpy as np
arr1=np.array([10, 20, 30, 40])
print(arr1)`

[10 20 30 40]

Print the type of the array

In [5]: `type(arr1)`

Out[5]: `numpy.ndarray`

Create a 2D Array

In [6]: `arr2=np.array([[1,2,3],[4,5,6]])
print(arr2)`

[[1 2 3]
 [4 5 6]]

Find - shape dimension size datatype

In [7]: `print(arr2.shape)
print(arr2.ndim)
print(arr2.size)
print(arr2.dtype)`

(2, 3)
2
6
int64

Create Zero Matrix

In [10]: `arr3=np.zeros((3,2))
print(arr3)`

[[0. 0.]
 [0. 0.]
 [0. 0.]]

In [11]: `arr3.dtype`

Out[11]: `dtype('float64')`

convert from float to int

In [15]: `arr3=arr3.astype(int)`

```
In [14]: print(arr3.dtype)
```

```
int64
```

Create a 2x4 array of ones

```
In [16]: arr4= np.ones((2,4))
print(arr4)
```

```
[[1. 1. 1. 1.]
 [1. 1. 1. 1.]]
```

```
In [17]: arr4=arr4.astype(int)
print(arr4)
```

```
[[1 1 1 1]
 [1 1 1 1]]
```

Create a 2x2 array filled with value 7

```
In [21]: arr5=np.full((2,2),7)
print(arr5)
```

```
[[7 7]
 [7 7]]
```

Generate 5 random numbers between 0 and 1

```
In [22]: print(np.random.rand(5)) #also in python to print random no. in b
```

```
[0.51779238 0.71834883 0.14522937 0.75505672 0.33963542]
```

Using arange Create numbers from 1 to 10

```
In [23]: print(np.arange(1,11))
```

```
[ 1  2  3  4  5  6  7  8  9 10]
```

Create even numbers between 10 and 20

```
In [24]: print(np.arange(10,21,2))
```

```
[10 12 14 16 18 20]
```

Create 5 evenly spaced values between 0 and 1

```
In [27]: print(np.linspace(0,1,5)) #for evenly spacing we can use linspace first write
```

```
[0.  0.25 0.5  0.75 1.  ]
```

Indexing

```
In [36]: arr1=np.array([[1,2,3], [4,5,6], [7,8,9]])
print(arr1)
```

```
[[1 2 3]
 [4 5 6]
 [7 8 9]]
```

```
In [37]: print(arr1[0:2,0:2])
```

```
[[1 2]
 [4 5]]
```

```
In [38]: print(arr1[0:2,1:])
```

```
[[2 3]
 [5 6]]
```

```
In [39]: print(arr1[1:,0:2])
```

```
[[4 5]
 [7 8]]
```

```
In [40]: print(arr1[1:,1:])
```

```
[[5 6]
 [8 9]]
```

```
In [41]: print(arr1[:,0])
```

```
[1 4 7]
```

```
In [42]: print(arr1[0,:])
```

```
[1 2 3]
```

Convert 1D to 2D Array

```
In [3]: arr1=np.array([1,2,3,4,5,6])
print(arr1)
arr1=np.array([1,2,3,4,5,6]).reshape(2,3)
print(arr1)
```

```
[1 2 3 4 5 6]
[[1 2 3]
 [4 5 6]]
```

Looping through array

```
In [5]: arr1=np.array([[1,2,3], [4,5,6], [7,8,9]])
for i in arr1:
    for j in i:
        print(j)
```

```
1
2
3
4
5
6
7
8
9
```

Add 10 to Each Element

```
In [6]: print(arr1+10) #this is known as broadcasting we are adding 10 to each el
```

```
[[11 12 13]
 [14 15 16]
 [17 18 19]]
```

Multiply Arrays

```
In [7]: a=np.array([[1,2], [3,4]])           #normal multiply it is multiply the e
        b=np.array([[1,2], [3,4]])
        print(a*b)
```

```
[[ 1  4]
 [ 9 16]]
```

```
In [8]: print(a@b)           #matrix multiplu see concept in notebook this we ha
```

```
[[ 7 10]
 [15 22]]
```

```
In [ ]:
```

Find mean, median, mode

```
In [ ]: arr = np.array([10,20,30,40])

        print(np.mean(arr))
        print(np.max(arr))
        print(np.min(arr))
```

```
In [14]: #find out mode
        !pip install scipy
```

Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: scipy in c:\programdata\anaconda3\lib\site-packages (1.16.3)
Requirement already satisfied: numpy<2.6,>=1.25.2 in c:\programdata\anaconda3\lib\site-packages (from scipy) (2.3.5)

```
In [15]: from scipy import stats
        arr1=np.array([10,10,10,20,30,40])
        print(stats.mode(arr1))
```

```
ModeResult(mode=np.int64(10), count=np.int64(3))
```

You have marks of students:

marks = [45, 78, 90, 66, 88, 72, 59] Tasks:

Convert to NumPy array

Find average marks

Find highest & lowest marks

Find students scoring above average

Add 5 grace marks to all students

```
In [9]: marks=np.array([45, 78, 90, 66, 88, 72, 59])
        print(marks)
        print("Average is",np.mean(marks))
        print("highest marks is",np.max(marks))
        print("lowest marks is",np.min(marks))
        print("add 5 grace marks is",marks+5)
```

```
[45 78 90 66 88 72 59]
Average is 71.14285714285714
highest marks is 90
lowest marks is 45
add 5 grace marks is [50 83 95 71 93 77 64]
```

```
In [17]: #students scoring marks above average
print(marks>np.mean(marks)) #this will give true and false but we want elements
print(marks[marks>np.mean(marks)])
```

```
[False True True False True True False]
[78 90 88 72]
```

Temperature recorded for 7 days:

temps = [30, 32, 35, 31, 29, 40, 38] Tasks:

Find days with temperature > 35

Count number of hot days

```
In [29]: temps=np.array([30, 32, 35, 31, 29, 40, 38])
print("days greater than temp 35 are", temps[temps>35])
print("no, of hot days are",len(temps))
```

```
days greater than temp 35 are [40 38]
no, of hot days are 7
```

Employee salaries: **(Homework questions)**

salary = [25000, 40000, 35000, 50000, 45000] Tasks:

Increase all salaries by 10%

Find employees earning > 40000

```
In [42]: salary=np.array([25000,40000,35000,50000,45000])
print("increased salaries by 10% are",salary*1.1 )
print("Employees earning more than 40000 are", salary[salary>40000])
```

```
increased salaries by 10% are [27500. 44000. 38500. 55000. 49500.]
Employees earning more than 40000 are [50000 45000]
```

A grayscale image (3x3 pixels):

img = [[100, 150, 200], [50, 75, 125], [0, 25, 255]] Tasks:

Convert to NumPy array

Increase brightness by 20

Clip values to max 255

```
In [55]: img=np.array([[100, 150, 200], [50, 75, 125], [0, 25, 255]])
print(img)
print("Increase brightness by 20 =",img+20)
print(np.clip(img,0,255)) #syntax of clipping=np.clip(array, min, max)
```

```
[[100 150 200]
 [ 50  75 125]
 [  0  25 255]]
Increase brightness by 20 = [[120 170 220]
 [ 70  95 145]
 [ 20  45 275]]
[[100 150 200]
 [ 50  75 125]
 [  0  25 255]]
```

Sales for 5 days:

sales = [200, 300, 250, 400, 350] Tasks:

Find total sales

Find day with highest sales

Find days where sales < 300

```
In [63]: sales=np.array([200, 300, 250, 400, 350])
print("total sales are", np.sum(sales))
print("sales less than 300 are", sales[sales<300])
```

total sales are 1500

sales less than 300 are [200 250]

Scores:

scores = [55, 65, 75, 85, 95] Tasks:

Get students with grades ≥ 75

Replace scores < 60 with 60

```
In [66]: scores=np.array( [55, 65, 75, 85, 95] )
print("students with grades more than equals to 75 are",scores[scores<300])
```

students with grades more than equals to 75 are [55 65 75 85 95]

homework questions of 31 march class python

```
In [74]: quote = "In this World nothing can be said to be Certain, Except Death and Taxes"

i=0

while i<len(quote):
    if quote[i].isupper():
        print(quote[i])
    i+=1    #this means i=i+1

len(quote) #this is to show you the length of quote including blanks
```

I
W
C
E
D
T

Out[74]: 72

Exercise 3: You have `people = ['Ana', 'Ben', 'Cara']` and `scores = [92, 85, 92]`. Use `zip()` to pair them and build a dict; if a name repeats, keep the higher score.

In []: